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Enhanced energy storage of aqueous zinc-carbon hybrid supercapacitors via employing alkaline medium and B, N dual doped carbon cathode



Lu Han^a, Xinlu Zhang^a, Junfeng Li^b, Hailong Huang^{a,*}, Xingtao Xu^c, Xinjuan Liu^d, Zhongli Yang^a, Min Xu^{a,*}, Likun Pan^{a,*}

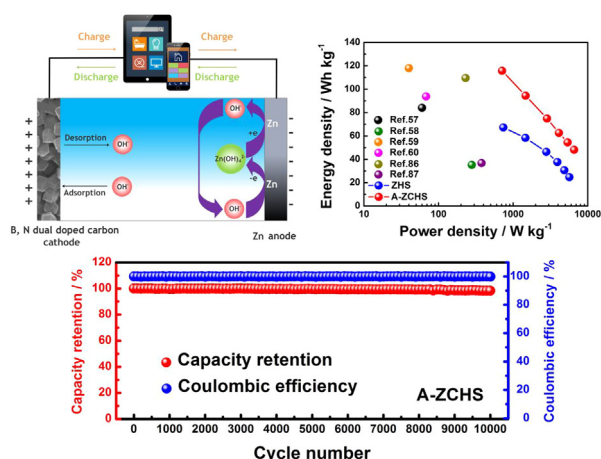
^a School of Physics and Electronic Science & Shanghai Key Laboratory of Magnetic Resonance, East China Normal University, Shanghai 200241, PR China

^b College of Logistics Engineering, Shanghai Maritime University, Shanghai 201306, PR China

^c International Center for Materials Nanoarchitectonics (WPI-MANA), National Institute for Materials Science, 1-1 Namiki, Tsukuba, Ibaraki 305-0044, Japan

^d Institute of Optoelectronic Materials and Devices, College of Optical and Electronic Technology, China Jiliang University, Hangzhou 310018, PR China

GRAPHICAL ABSTRACT



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ABSTRACT

Zinc-based energy storage systems (zinc-air, zinc-nickel and zinc-ion batteries and zinc-ion hybrid supercapacitors (ZHSSs) are considered as promising power sources for wide applications from personal electronic devices to electric vehicles. However, these systems, especially the Zn-based hybrid supercapacitors, display unsatisfying power density and energy density, which should be enhanced for their large-scale applications. In this work, aqueous alkaline zinc-carbon hybrid supercapacitors (A-ZCHS) were designed, consisting of B, N dual doped carbon cathode, Zn anode and KOH electrolyte. The B, N dual doped carbon was prepared via thermal treatment of metal-organic frameworks and boric acid, which exhibits abundant hierarchical pore structure (micropore, mesopore and macropore) and suitable defect construction, promoting ion diffusion/charge transfer and providing more rapid surface pseudocapacitance reaction. More obviously, when the optimized B, N dual doped carbon was used as cathode in A-ZCHS and ZHS, more capacitive charge storage and rapid electrochemical kinetics can be observed in A-ZCHS than in ZHS. Therefore, the optimized A-ZCHS displays a high energy density of 115.7 Wh kg⁻¹ at the power density of 711.6 W kg⁻¹ with excellent stability, which is much better than

* Corresponding authors.

E-mail addresses: huanghao3310774@163.com (H. Huang), xumin@phy.ecnu.edu.cn (M. Xu), lkpan@phy.ecnu.edu.cn (L. Pan).

most of ZHSs reported previously. The A-ZCHS should be a promising candidate for energy storage applications.

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1. Introduction

The depletion of fossil fuels and growing energy demands have urgently called for exploitation and utilization of some renewable/clean energy such as water power, wind power, and solar energy and wave energy [1–20]. However, limited by time and space, the extensive use of renewable/clean energy is intermittent, which is still unsatisfying for the pattern of practical use and energy demand [21–23]. Rechargeable electrochemical energy storage (rechargeable batteries and supercapacitors) [24–44], as the main form of intermittent renewable/clean energy storage, can realize the conversion between electrical energy and chemical energy. Considering the features of zinc metal in low flammability, low cost, high stability and high abundance, zinc-based energy storage systems (zinc-air, zinc-nickel and zinc-ion batteries and zinc-ion hybrid supercapacitors (ZHS)) are considered as promising power sources for wide and practical applications [45–56]. However, plagued by their poor cycling stability and unsatisfying energy storage, the electrochemical performance of these devices is still needs to be promoted to meet practical use and energy demand.

Recently, ZHS, typically zinc-carbon based hybrid supercapacitor, is considered as a new-type of hybrid supercapacitors and zinc-based energy storage systems, holding great promise in realizing superior energy storage devices. Dong et al. assembled activated carbon (AC) cathode //ZnSO₄ (aq)//Zn anode ZHS, exhibiting an energy density of 84 Wh kg^{−1} at about 60 W kg^{−1} [57]. Wang et al. reported that MXene-reduced graphene oxide aerogel achieved an energy density of 34.9 Wh kg^{−1} at 279.9 W kg^{−1} when evaluated as the cathode for ZHS [58]. Yu et al. fabricated hierarchically porous carbon as the cathode of ZHS, displaying an energy density of 118 Wh kg^{−1} at about 40 W kg^{−1} [59]. He et al. prepared AC carbonized by corncob and when it was used as the cathode of ZHS, an energy density of 94 Wh kg^{−1} at 68 W kg^{−1} was achieved [60]. In these reported ZHSs, ZnSO₄ aqueous electrolyte is used as the electrolyte and the electrical double layer capacitance primarily comes from anion physical adsorbing on/desorbing from the surface of carbon-based cathode (carbon composites or carbon-based materials). Meanwhile the faradaic contribution is provided through Zn (Zn²⁺) depositing on/stripping from Zn foil anode during charge/discharge process. Despite some progress has been made for currently reported ZHSs, their unsatisfying power density and energy density still need to be enhanced for wide and practical applications.

For the ZHS, the physical adsorption/desorption process of carbon-based cathode, influenced by the conductivity of ZnSO₄ aqueous electrolyte, plays a crucial effect on its power density and energy density. 2 M KOH aqueous alkaline electrolyte (344 mS cm^{−1}) is found to possess higher conductivity than 2 M ZnSO₄ aqueous electrolyte (49.4 mS cm^{−1}) as shown in Table S1, which means faster ion migration rate in aqueous alkaline electrolyte. Meanwhile alkaline environment also can provide stable faradaic oxidation/reduction reactions on the battery-type Zn anode ($\text{Zn} + 4\text{OH}^- - 2\text{e}^- = \text{Zn}(\text{OH})_4^{2-}$, $E^0 = -1.199$ V versus SHE, standard hydrogen electrode) [61]. Therefore, zinc-carbon hybrid supercapacitors using KOH alkaline electrolyte should exhibit higher power density and energy density than those using ZnSO₄ aqueous electrolyte. Unfortunately, the zinc-carbon hybrid supercapacitors

operated in aqueous alkaline environment still have been rarely explored by now.

In this work, inspired by the construction of aqueous alkaline zinc-nickel batteries and asymmetric hybrid supercapacitors [61–66], aqueous alkaline zinc-carbon hybrid supercapacitors (A-ZCHS) were designed, consisting of B, N dual doped carbon cathode, Zn anode and KOH electrolyte, with outstanding energy storage performance. Notably in aqueous alkaline electrolyte, optimizing the structure of carbon-based cathode materials is crucial for facilitating the physical adsorption/desorption process, as reported previously for ZHSs [57–60]. Therefore, B, N dual doped carbon was used here instead of conventionally reported AC cathode. The energy storage mechanism in A-ZCHS is illustrated in Fig. 1. Significantly, the stored electrical energy of A-ZCHS corresponds to the amount of adsorbed ions on the surface of carbon-based cathode, which means that regulating the carbon frameworks with porous structure and changing the electronic structure of carbon surface via doping heteroatoms should be effective ways to enhance the electrochemical performance of A-ZCHS. Herein, hierarchically porous B, N dual doped carbon was prepared via thermally annealing metal-organic frameworks (MOFs) and boric acid. Benefiting from rich porous structure of MOFs, the optimized B, N dual doped carbon exhibits abundant porosity and high specific surface area. Meanwhile, the density state and electronic structure of carbon surface can be significantly changed by B, N dual doping, thus enhancing its capacitance [67–73]. When the optimized B, N dual doped carbon was used as cathode, the A-ZCHS can work in a voltage range of 0.2–1.8 V and display a high energy density of 115.7 Wh kg^{−1} at the power density of 711.6 W kg^{−1}, which is better than those of ZHS. Importantly, a capacity retention of 99.2% can be achieved after 10,000 cycles, showing excellent cycling stability.

2. Experimental

2.1. Preparation of Zn-containing MOFs (ZIF-8)

ZIF-8 was synthesized in a typical process. Firstly, 2-methylimidazole (2.63 g) and Zn(NO₃)₂ (2.58 g) were completely dispersed in methanol solutions (200 mL), respectively. Subsequently, Zn(NO₃)₂ methanol solution was dropped into 2-methylimidazole methanol solution, which was maintained at static settlement for 24 h. And then, white powders were collected by centrifugation, and washed carefully with methanol for about 5 times. The obtained ZIF-8 was dried in a vacuum oven at 60 °C for 48 h.

2.2. Preparation of B, N dual doped carbon

Typically, as shown in Fig. 2, the resulting ZIF-8 (100 mg) was firstly mingled with boric acid (0 mg, 10 mg, 30 mg and 50 mg). And then, the obtained mixtures were annealed in tube furnace at 800 °C under N₂ atmosphere for about 2 h, with a heating rate of 2 °C min^{−1}. The black powders were named as N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5 according to the amount of boric acid (0 mg, 10 mg, 30 mg and 50 mg). Finally, all the samples were washed with 2 M HCl aqueous solution and dried at 60 °C in a vacuum oven for about 48 h.

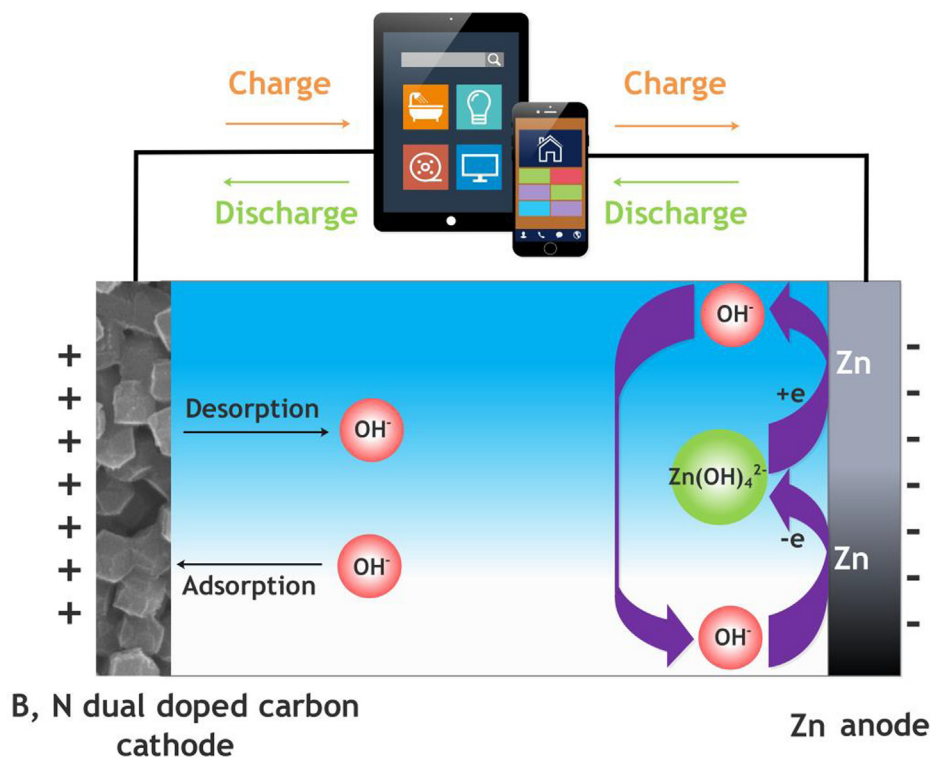


Fig. 1. Schematic illustration of energy storage mechanism in A-ZCHS.

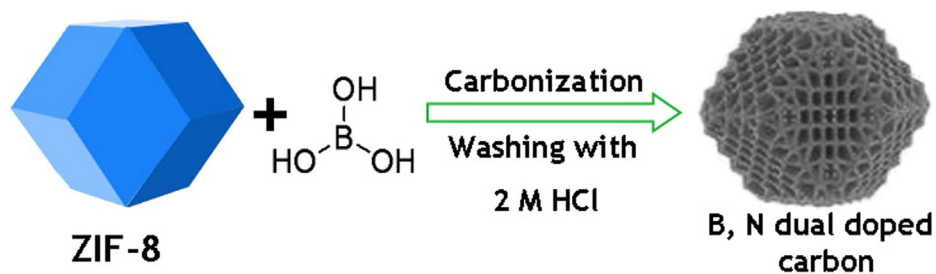


Fig. 2. Preparation process of B, N dual doped carbon.

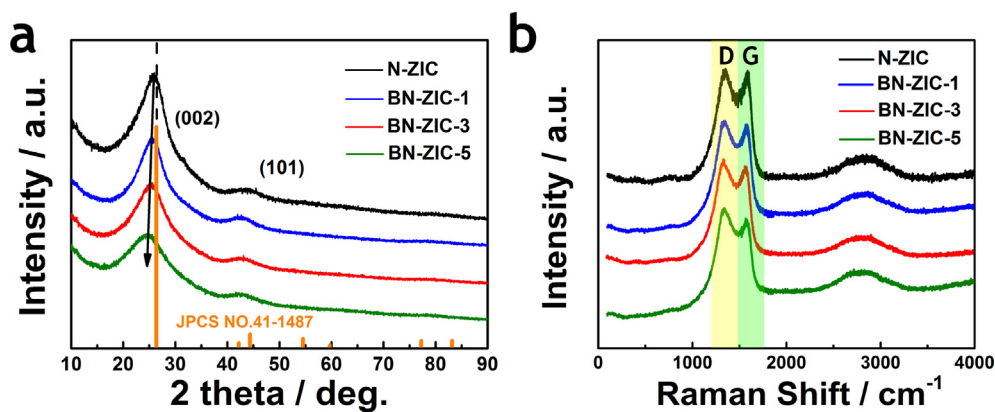


Fig. 3. (a) XRD patterns and (b) Raman spectra of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5.

The material characterizations and electrochemical measurements have been described detailedly in ESI†.

3. Results and discussion

As shown in Fig. 3a, X-ray powder diffraction (XRD) measurement examined the crystallographic structures of all samples. In all of XRD patterns, two broad characteristic humps can be recognized as (002) and (101) planes of carbon (JPCS no.41-1487) at around 26° and 43° respectively. Obviously, the peak around 26° shifts slowly toward lower 2θ values with increasing amount of boric acid during the sample preparation, since B, N dual doping can expand the interlayer spacing of carbon [74]. From the transmission electron microscopy (TEM) images of BN-ZIC-3 in Fig. S2, the identifiable lattice spacing of BN-ZIC-3 is measured to be 0.38 nm, which is larger than 0.34 nm (JPCS no.41-1487) and fits the (002) plane of carbon. Besides, as shown in Table S2, the (002) plane in full width at half maxima (FWHM) becomes broader with the increasing amount of boric acid, indicating the enhanced defects caused by the incorporation of heteroatoms (B, N dual doping) and some small stacks in sp^2 -bonded conjugated graphitic carbons [68]. Furthermore, in Fig. 3b, the Raman spectra were used to characterize the disorder and graphitization degree of all samples. D band (1350 cm^{-1}) and G band (1580 cm^{-1}) are associated with the vibration of sp^3 carbon atoms in the defects/disorders and sp^2 carbon atoms in the graphitic 2D hexagonal lattice, respectively. The D band to G band (I_D/I_G) intensity ratio is considered as a useful indicator to evaluate the disorders of carbon material. According to Table S2, the I_D/I_G intensity ratios of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5 are 1.01, 1.03, 1.05 and 1.09, respectively, revealing more defects and disorders in carbon frameworks attributed to B, N dual doping with increasing amount of boric acid during the preparation [75]. The recent mechanisms underlying electrode reactions prove that suitable defect construction in electrode materials via exposing coordination unsaturated sites can effectively promote the charge transfer and ion diffusion. Meanwhile, those exposed defects can provide more active adsorption/storage sites for intermediate species and guest ions, thus improving the electrochemical performance [76,77].

The structures and morphologies of all samples were investigated by scanning electron microscopy (SEM). In Fig. 4a–c, N-ZIC, BN-ZIC-1 and BN-ZIC-3 exhibit analogical shape to their MOFs precursor (Fig. S3), indicating that their structures are well inherited from original morphology of MOFs even after the addition of suit-

able amount of boric acid and following carbonization process. However, BN-ZIC-5 exhibits a collapsed framework structure, as shown in Fig. 4d, mainly due to the addition of excess boric acid during the preparation. In Fig. 4e, the porosities of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5 were investigated by the N_2 adsorption/desorption isotherms. All isotherms display hybrid type I-IV curves with two obvious sharp rises ($P/P_0 < 0.15$, P/P_0 greater than 0.9) and small hysteresis loops ($0.30 < P/P_0 < 0.9$), which means the co-existence of micropore, mesopore and macropore [78–80]. It should be noted that the isotherms of N-ZIC, BN-ZIC-1 and BN-ZIC-3 exhibit a steep uptake in low-pressure area ($P/P_0 < 0.15$, typical for microporous materials), but a more slow rise in the isotherm of BN-ZIC-5 means that the proportion of the microporous structure decreases, due to the addition of excess boric acid during the preparation. Furthermore, the pore size distributions of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5 can be seen in Fig. S1. Obviously, with the increasing amount of boric acid during the sample preparation, higher mesopore proportion is obtained in these MOFs-derived carbon frameworks. In addition, as shown in Fig. S1d, when excess boric acid was introduced during the sample preparation, a higher mesopore proportion and collapsed carbon framework structure was obtained, consistent to the SEM observation in Fig. 4d. The pore characteristics and Brunauer-Emmett-Teller specific surface areas of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5 are listed in Table S2. Clearly, BN-ZIC-3 displays a higher specific surface area of $926.67\text{ m}^2\cdot\text{g}^{-1}$ than N-ZIC ($574.48\text{ m}^2\cdot\text{g}^{-1}$), BN-ZIC-1 ($802.74\text{ m}^2\cdot\text{g}^{-1}$) and BN-ZIC-5 ($197.15\text{ m}^2\cdot\text{g}^{-1}$). This is mainly due to the remained similar structures and morphologies of MOFs precursor with rich porosity in BN-ZIC-3, and the TEM image of BN-ZIC-3 in Fig. S2 also proves this point. Meanwhile, adding suitable amount of boric acid during the preparation of BN-ZIC-3 could guarantee its abundant and hierarchical pore structure consisting of micropore, mesopore and macropore, and avoid its structural collapse, which can enhance the mass transport and ion diffusion, ensuring the excellent electrochemical performance [81,82].

X-ray photoelectron spectroscopy (XPS) was performed to analyze the chemical composition of all samples, and the results are shown in Table S3, which clearly indicate the successful B and N dual doping in BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5. And their B contents are 2.84 at.%, 3.82 at.% and 7.24 at.% respectively. Interestingly, with the enhanced B doping in the samples, N and O contents also increase obviously, but the reason is still to be investigated. As seen from Fig. 5a, the energy dispersive X-ray spec-

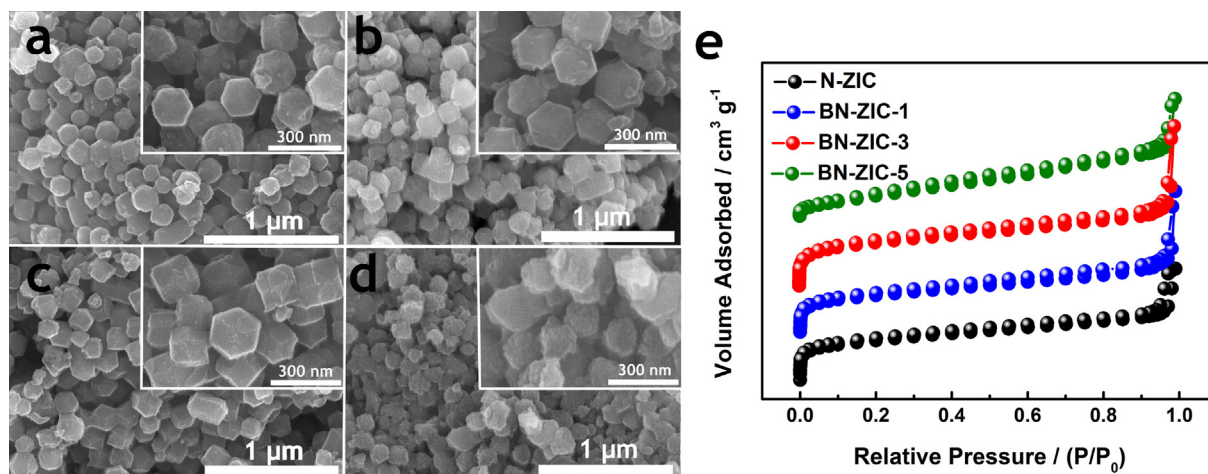


Fig. 4. (a–d) SEM images and (e) nitrogen adsorption–desorption isotherms of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5.

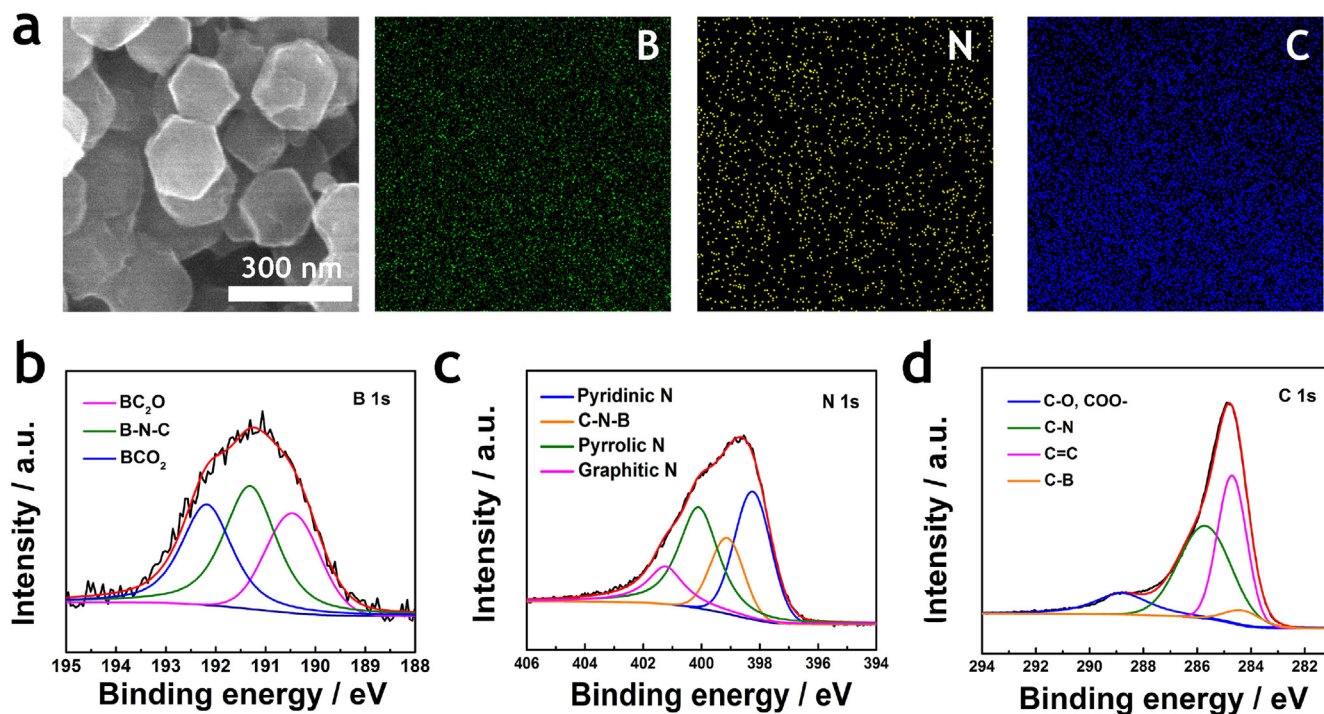


Fig. 5. EDS elemental mapping images and XPS (a) B 1 s, (b) N 1 s and (c) C 1 s spectra of BN-ZIC-3.

troscopy (EDS) elemental mappings of BN-ZIC-3 further demonstrate the successful doping of B and N elements. In addition, the chemical states of B, N and C in BN-ZIC-3 are further displayed in Fig. 5b–d. As shown in Fig. 5b, three peaks centered at about 190.4, 191.3, and 192.2 eV can be identified as BC_2O , B–N–C and BCO_2 bonds [70], respectively, indicating that B doping exists in the forms of boron-carbon-oxygen group species (BC_2O and BCO_2) and B–N–C bonds in BN-ZIC-3. The XPS N 1 s spectrum (Fig. 5c) of BN-ZIC-3 possesses four fitting peaks at about 398.2 eV (pyridinic N), 399.2 eV (C–N–B), 400.1 eV (pyrrolic N) and 401 eV (graphitic N) [70]. In Fig. 5d, the C1s spectrum of BN-ZIC-3 gives four peaks representing B–C (284.3 eV), C=C (284.7 eV), C–N (286 eV) and C–O (COO–) (288.8 eV) bonds [83]. Therefore, B, N dual doping is introduced in BN-ZIC-3 in the forms of B-containing groups (BC_2O and BCO_2 species), N-containing groups (pyridinic N, pyrrolic N and quaternary N species) and C–N–B species in carbon framework. These surface defects can enhance pseudocapacitance contribution, which benefits the electrochemical performance [68,70].

In order to investigate the electrochemical performances of the samples, a series of electrochemical tests have been measured in 2 M KOH aqueous electrolyte under three-electrode configuration. Fig. 6a shows the results of electrochemical impedance spectroscopy (EIS). Obviously, each Nyquist plot has a steep slope (low-frequency region) and a small semicircle (high frequency region), which are related to Warburg impedance and charge-transfer resistance (R_{ct}) [83], respectively. Fitting the EIS spectra based on the equivalent circuit model (in the inset of Fig. 6a) can evaluate the R_{ct} values. According to Table S2, the fitted R_{ct} values of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5 are 0.962, 0.421, 0.223 and 1.28 Ω , respectively. Clearly, BN-ZIC-3 possesses a smaller R_{ct} than N-ZIC, BN-ZIC-1 and BN-ZIC-5, reflecting its good interfacial charge transfer between electrode material and electrolyte, possibly because suitable defect construction caused by B, N dual doping in BN-ZIC-3 effectively promotes the interfacial charge transfer process [72]. The low R_{ct} can lead to fast pseudocapacitance

reaction and superior capacitive performance [68]. Furthermore, the slope line in low frequency region of Nyquist plots is contributed by the ion diffusion. Compared with N-ZIC, BN-ZIC-1 and BN-ZIC-5, a much steeper line gradient of BN-ZIC-3 corresponds to faster ion diffusion, representing a more ideal capacitor. This is due to the high specific surface and hierarchical pore structure of BN-ZIC-3, facilitating the ion diffusion at the interface between electrolyte and electrode [68]. The cyclic voltammetry (CV) and galvanostatic charge-discharge (GCD) tests were investigated to further evaluate the electrochemical properties of the samples. In Fig. 6b, All CV curves display similar distorted rectangular shapes with some obvious reversible redox peaks caused by surface functional groups, suggesting the coexistence of electrical double layer (EDL) capacitive behavior and surface pseudocapacitance behavior. Additionally, larger integral area in the CV curve of BN-ZIC-3 could be obtained, compared with those for N-ZIC, BN-ZIC-1 and BN-ZIC-5, because of the enhanced ionic migration caused by the more suitable hierarchical pore structure and optimized surface defect construction caused by B, N dual doping, indicating more excellent electrochemical performance of BN-ZIC-3. The corresponding reversible capacities were evaluated and compared by GCD curves (Fig. 6c), showing that BN-ZIC-3 exhibits a higher reversible capacity at 1 A g^{-1} . As shown in Fig. 6d, BN-ZIC-3 exhibits higher capacities than N-ZIC, BN-ZIC-1 and BN-ZIC-5 at all current densities, which are calculated to be 108.6, 90.1, 72.6, 60.5, 52.3 and 46.3 mAh g^{-1} at current densities of 1, 2, 4, 6, 8 and 10 A g^{-1} (Table 1), respectively.

In order to further evaluate the practical applications of the devices in different electrolytes, A-ZCHS and ZHS were assembled by Zn foil as anode and BN-ZIC-3 as cathode with 2 M KOH and 2 M ZnSO_4 as aqueous electrolytes, respectively. As shown in Fig. 7a, the CV curves of A-ZCHS in a voltage window of 0.2–1.8 V at various scan rates (2–10 mV s^{-1}) are recorded and similar shapes of all the CV curves can be observed, when the scan rate increases, indicating its outstanding reversible stability. The capacities of A-ZCHS were calculated via GCD measurements (Fig. 7b) at

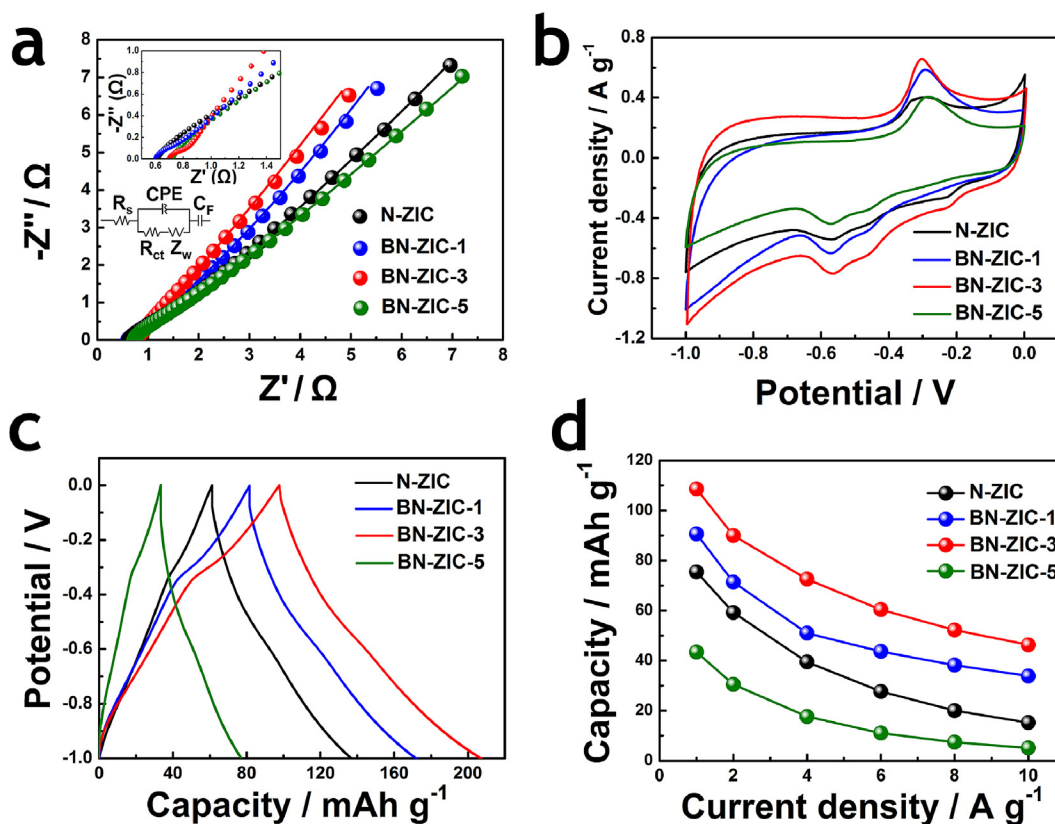


Fig. 6. (a) Nyquist plots of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5. Inset shows the corresponding magnified spectra in the high-frequency region and the equivalent circuit model. (b) CV curves at 2 mV s^{-1} and (c) GCD curves at 1 A g^{-1} of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5. (d) Specific capacitances of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5 as a function of discharge current density.

Table 1

Discharge specific capacities of N-ZIC, BN-ZIC-1, BN-ZIC-3 and BN-ZIC-5 in 2 M KOH aqueous electrolyte at different current densities ($1\text{--}10 \text{ A g}^{-1}$) in a three-electrode configuration.

Current (A g^{-1})	1	2	4	6	8	10
N-ZIC/Capacity (mAh g^{-1})	75.52	59.27	39.48	27.72	20.07	15.25
BN-ZIC-1/Capacity (mAh g^{-1})	90.69	71.43	51.02	43.67	38.2	33.89
BN-ZIC-3/Capacity (mAh g^{-1})	108.64	90.04	72.58	60.45	52.27	46.33
BN-ZIC-5/Capacity (mAh g^{-1})	43.44	30.52	17.66	11.08	7.44	5.11

various current densities ($2\text{--}10 \text{ A g}^{-1}$) and the corresponding results can be seen in Fig. 7c. The A-ZCHS displays a maximum capacity of 162.6 mAh g^{-1} at 1 A g^{-1} . Correspondingly, the CV and GCD curves of ZHS were also measured, as shown in Fig. 7d and e. Its calculated maximum capacity is 92.7 mAh g^{-1} at 1 A g^{-1} (Fig. 7f), much lower than that of A-ZCHS. This should be attributed to faster ion migration in aqueous alkaline electrolyte, resulting in faster physical adsorption/desorption process of carbon-based cathode. Therefore, BN-ZIC-3 based A-ZCHS exhibits higher capacities than ZHS as well as N-ZIC, BN-ZIC-1 and BN-ZIC-5 based A-ZCHS (Table S4) at all current densities, which are calculated to be 162.6 , 128.5 , 105.1 , 91.1 , 80.9 and 73.1 mAh g^{-1} at current densities of 1 , 2 , 4 , 6 , 8 and 10 A g^{-1} . The related Ragone plots of BN-ZIC-3 based A-ZCHS and ZHS are shown in Fig. 7g. Notably, A-ZCHS reveals a maximum energy density of 115.7 Wh kg^{-1} at the power density of 711.6 W kg^{-1} , showing better electrochemical performance than those of ZHS in this work and most previously reported ZHSs (Table S5). The continuous charge-discharge test at 10 A g^{-1} for 10,000 cycles was measured to investigate the long cycling stability of A-ZCHS. The corresponding results in Fig. 7h indicate its outstanding cycling stability with a capacitance reten-

tion of 99.2% and columbic efficiency of nearly 100% after 10,000 charge-discharge cycles tests.

Based on the above analysis, the A-ZCHS possesses outstanding electrochemical performance. The kinetic behaviors of A-ZCHS and ZHS were further investigated based on the relationship between cathodic/anodic peak current (i_p) and scan rate (v). According to the CV curves of A-ZCHS and ZHS at different scan rates ($2\text{--}10 \text{ mV s}^{-1}$) in Fig. 8a and d, the linear relations of peak current and scan rate were analyzed by plotting $\log(i)$ versus $\log(v)$ [84,85]. A power-law relationship of current (i , A) and scan rate (v , mV s^{-1}) obeys the following equations:

$$i = av^b \quad (1)$$

$$\log(i) = b \log(v) + \log(a) \quad (2)$$

In the above equations, a and b are constants. The slope obtained via linear fitting of $\log(i)$ and $\log(v)$ can be used to calculate b value. When $b = 0.5$, the kinetic behavior can be considered as a diffusion-limited charge/discharge process, while $b = 1$ represents a capacitive process (EDL capacitors) [85]. In Fig. 8b, the b values for cathodic (b_1) and anodic (b_2) peak currents of

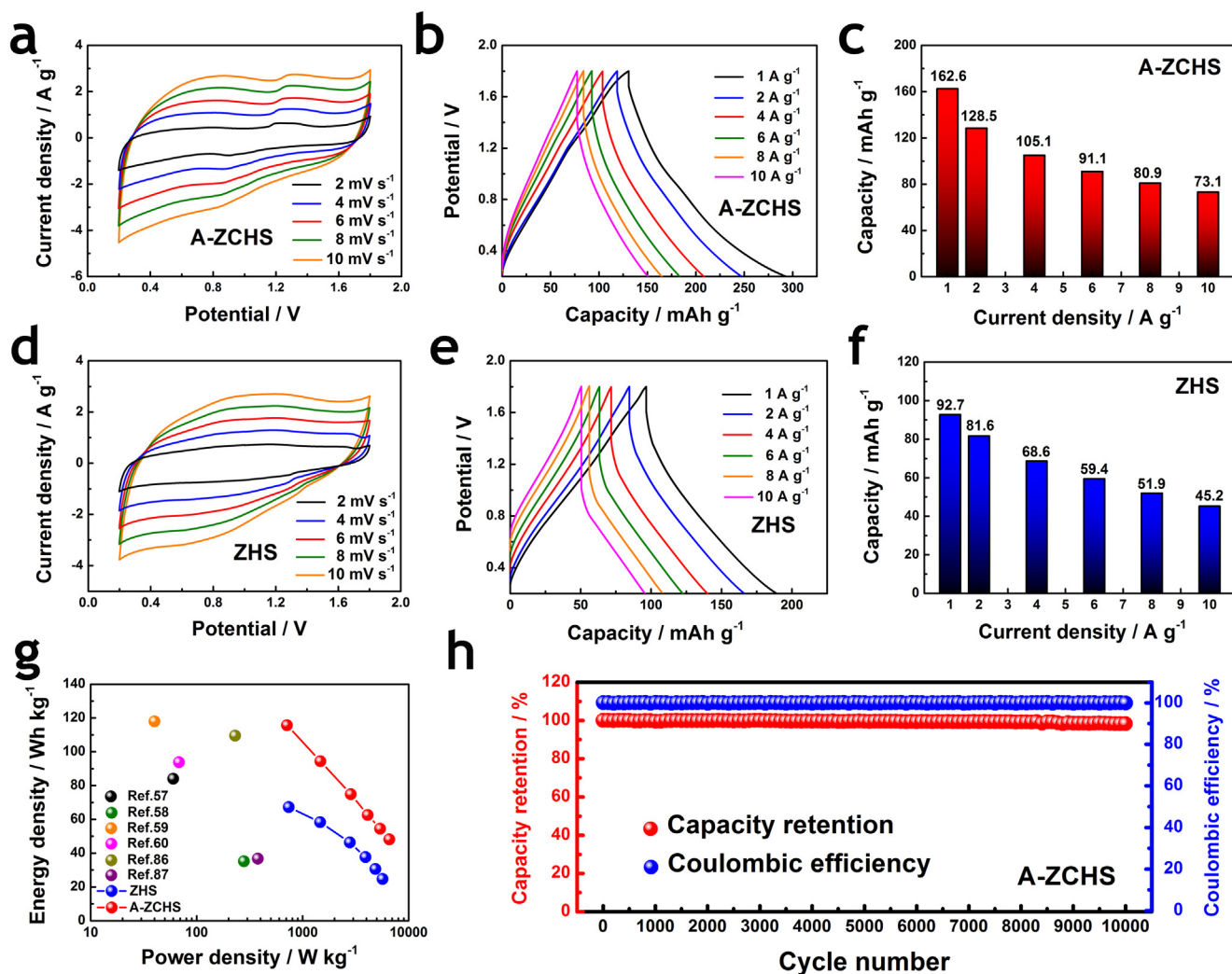


Fig. 7. Electrochemical behaviors of A-ZCHS and ZHS: (a) CV curves at 2–10 mV s⁻¹ and (b) GCD curves at 2–10 A g⁻¹ (0.2–1.8 V) of A-ZCHS. (c) Capacities of A-ZCHS at various current densities. (d) CV curves at 2–10 mV s⁻¹ and (e) GCD curves at 2–10 A g⁻¹ (0.2–1.8 V) of ZHS. (f) Capacities of ZHS at various current densities. (g) Ragone plots of A-ZCHS and ZHS. (h) Cycling performance of A-ZCHS at a current density of 10 A g⁻¹. (See above-mentioned references for further information.)

A-ZCHS are 0.8922 and 0.8711, while in ZHS the b values for cathodic (b_3) and anodic (b_4) peak currents are 0.7992 and 0.8388 (Fig. 8e), respectively. The results demonstrate that both of diffusion-controlled and surface pseudocapacitance behaviors are responsible for the charge/discharge processes in both A-ZCHS and ZHS. The diffusion-controlled contribution is mainly due to the abundant and suitable hierarchical pore structure consisting of micropore, mesopore and macropore. Meanwhile the capacitive contribution is mainly ascribed to rapid surface pseudocapacitance reaction caused by suitable defect construction (B, N dual doping) in carbon cathode. But more obviously, b values in A-ZCHS are higher than those in ZHS ($b_1 > b_3$ and $b_2 > b_4$), which means high capacitive contribution in A-ZCHS than ZHS due to faster ion migration rate and more rapid surface pseudocapacitance reaction on the carbon cathode surface in aqueous alkaline electrolyte, leading to faster interfacial charge transfer. In addition, the capacitive contribution ratios can be quantified by the following equation:

$$i_{(v)} = k_1 v + k_2 v^{1/2} \quad (3)$$

The capacitive and diffusion-controlled contribution ratios of A-ZCHS and ZHS at various scan rates are displayed in Fig. 8c and f. The capacitive contribution ratios of A-ZCHS were calculated to

be 69.8%, 76.6%, 80%, 82.2% and 83.6%, at scan rates of 2, 4, 6, 8 and 10 mV s⁻¹, respectively, which are higher than those of ZHS. The results further indicate that more capacitive charge storage and faster electrochemical kinetics happen on the carbon cathode surface in A-ZCHS than in ZHS at all scan rates (2–10 mV s⁻¹), exhibiting more excellent electrochemical performance in 2 M KOH aqueous electrolytes.

4. Conclusions

In conclusion, A-ZCHS, as one of zinc-based energy storage systems, was designed and assembled by B, N dual doped carbon cathode and Zn anode in 2 M KOH aqueous electrolyte. The carbon cathode was prepared from Zn-containing MOFs and regulated by B, N dual doping. The optimized B, N dual doped carbon retains similar morphology of MOFs precursor and exhibits abundant hierarchical pore structure (micropore, mesopore and macropore), facilitating ion diffusion and enhancing mass transport. Meanwhile, B, N dual doping can significantly influence electronic structure and density state of carbon surface, thus enhancing pseudocapacitance contribution. When the optimized B, N dual doped carbon was used as cathode in A-ZCHS and ZHS, more capacitive contribution/capacitive charge storage and rapider elec-

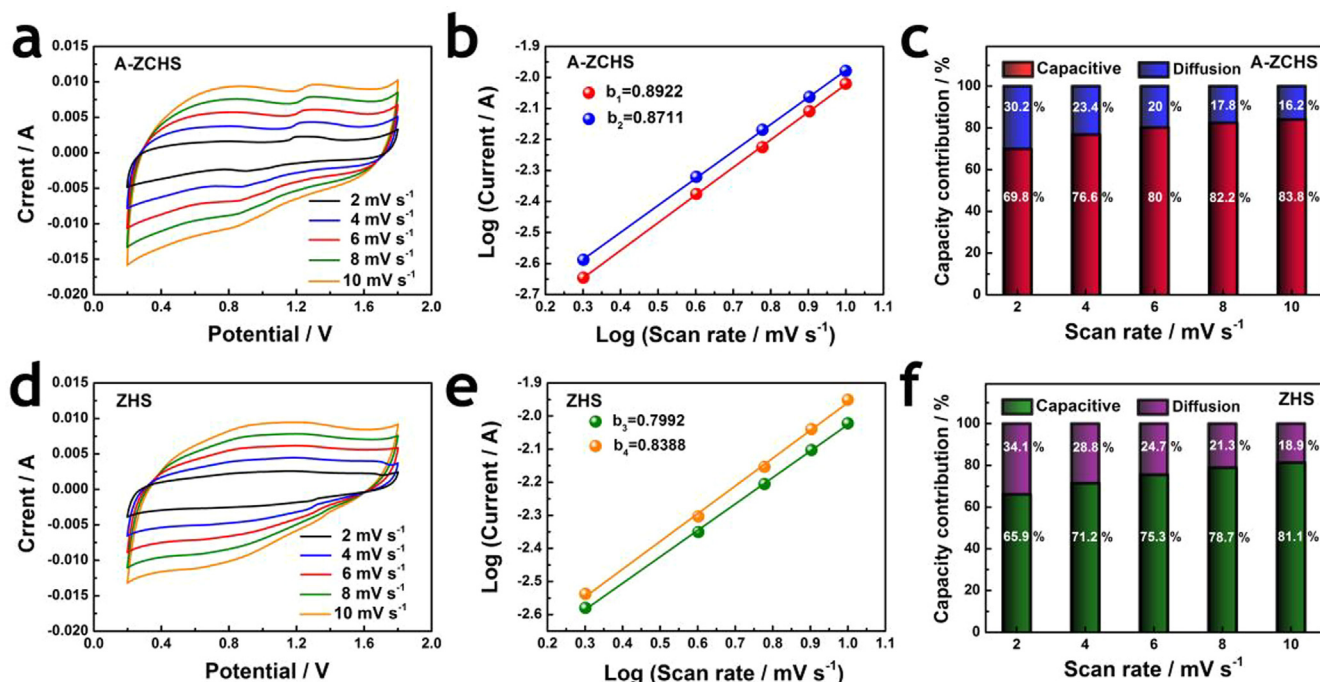


Fig. 8. (a, d) CV curves of A-ZCHS and ZHS at different scan rates ranging from 2 to 10 mV s⁻¹. (b, e) Corresponding log (*i*) versus log (*v*) plots of A-ZCHS and ZHS. (c, f) Normalized capacity contributions of A-ZCHS and ZHS at different scan rates from capacitive and diffusion-controlled behaviors.

trochemical kinetics happen in A-ZCHS than in ZHS, mainly due to faster ion migration rate and rapid surface pseudocapacitance reaction at the carbon cathode surface in aqueous alkaline electrolyte. Therefore, A-ZCHS displays a high energy density of 115.7 Wh kg⁻¹ at 711.6 W kg⁻¹ and excellent cycling stability (99.2% capacity retention) after 10,000 cycles. The strategy in this work via exploring alkaline medium and structural surface regulation of carbon cathode material should effectively promote the energy storage performance of A-ZCHS, offering a meaningful guidance in aqueous Zn-based energy storage field.

CRediT authorship contribution statement

Lu Han: Conceptualization, Methodology, Writing - original draft. **Xinlu Zhang:** Visualization. **Junfeng Li:** Validation. **Hailong Huang:** Methodology, Validation. **Xingtao Xu:** Data curation. **Xinjuan Liu:** Investigation. **Zhongli Yang:** Formal analysis. **Min Xu:** Resources, Supervision. **Likun Pan:** Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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